

S&DS 365 / 665
Intermediate Machine Learning
Sparsity and Graphs

February 25

Yale

Reminders

- Assignment 2 due this Thursday (midnight)
- Midterm next Wednesday in class
- Assignment 3 released after midterm

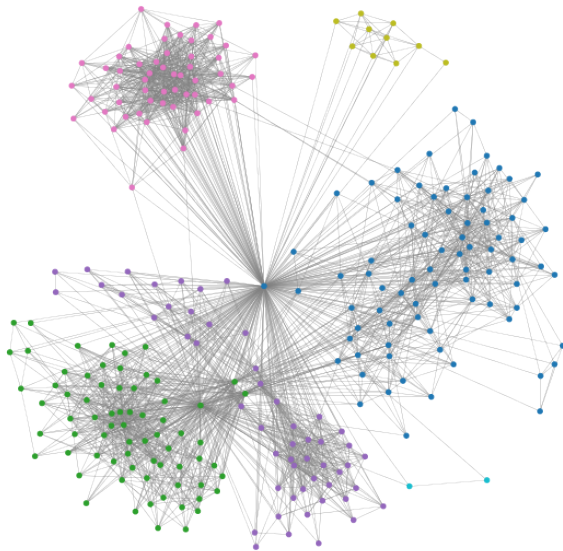
Outline

For today:

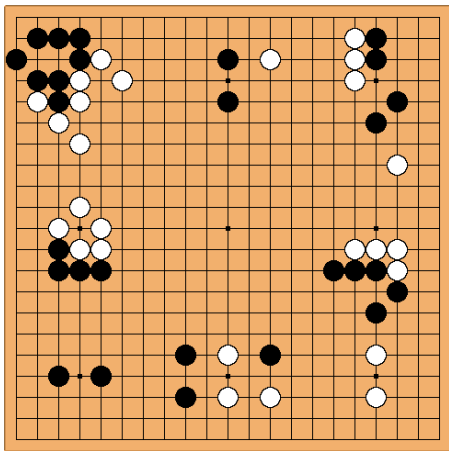
- Graphs of data/distributions

Graphs

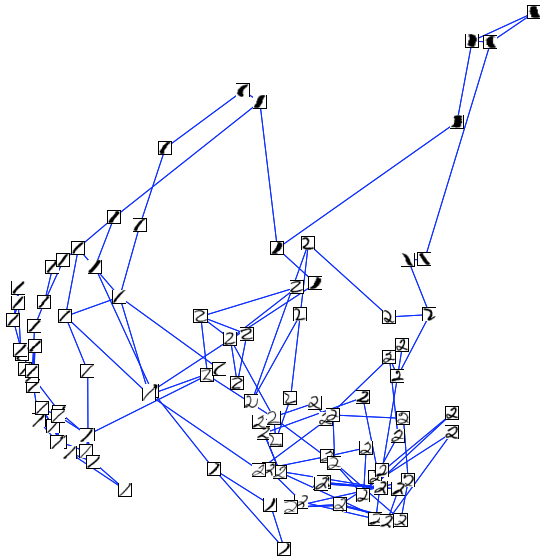
- Next two lectures touch on ML for data having graphical structure
- This is quite common



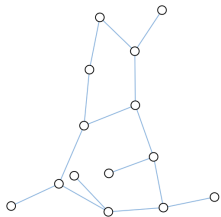
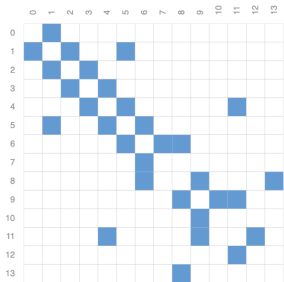
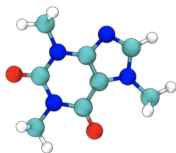
social networks (Facebook friends graph)



games

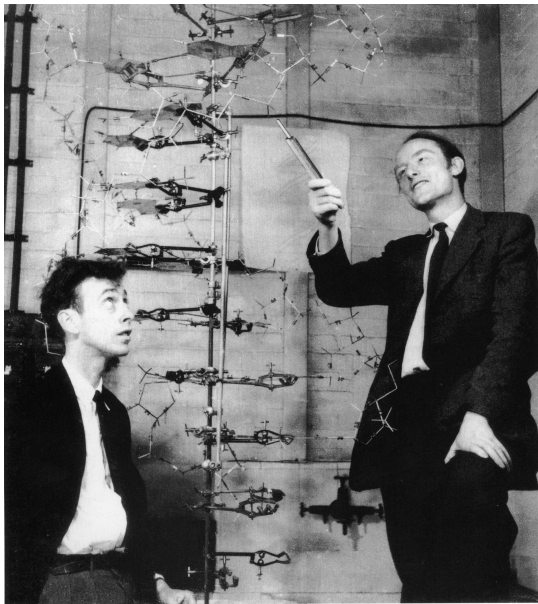


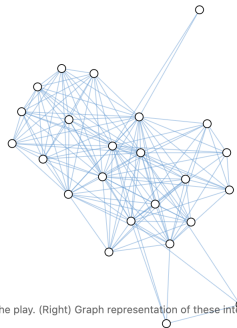
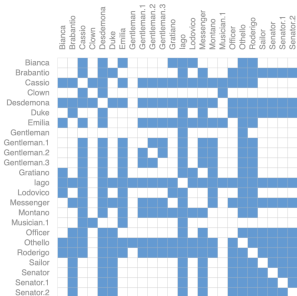
semi-supervised learning



(Left) 3d representation of the Caffeine molecule (Center) Adjacency matrix of the bonds in the molecule (Right) Graph representation of the molecule.

<https://distill.pub/2021/gnn-intro/>

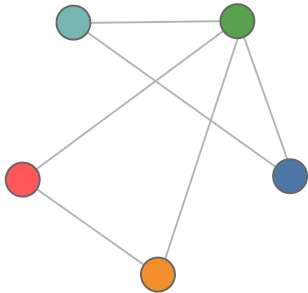




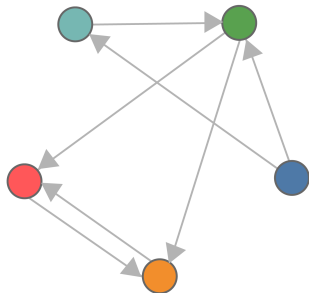
(Left) Image of a scene from the play "Othello". (Center) Adjacency matrix of the interaction between characters in the play. (Right) Graph representation of these interactions.

<https://distill.pub/2021/gnn-intro/>

Undirected graph



Directed graph





Graphs

- A natural language for describing various data
- Give information about relationships between variables
- Associated with each multivariate distribution

Undirected Graphs

A graph $G = (V, E)$ has vertices V , edges E .

If $X = (X_1, \dots, X_p)$ is a random variable, we will study graphs where there are p vertices, one for each X_j .

The graph will encode conditional independence relations among the variables.

Example: Gaussian data

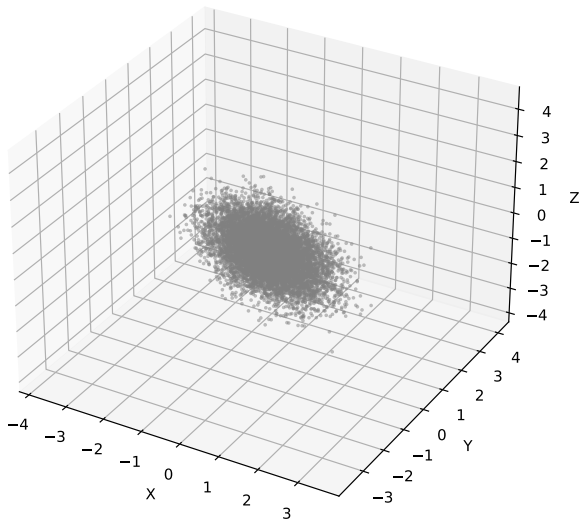
We have a three-dimensional Gaussian (X, Y, Z) with covariance

$$\Sigma = \begin{pmatrix} 3.1 & -2.4 & 2.1 \\ -2.4 & 2.6 & -2.4 \\ 2.1 & -2.4 & 3.1 \end{pmatrix}$$

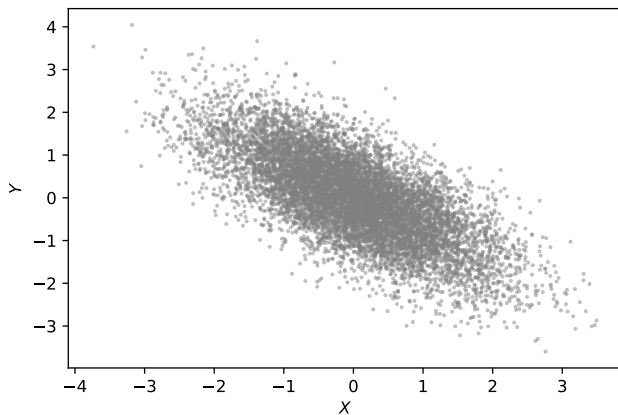
So, all pairs are correlated

Example: Gaussian data

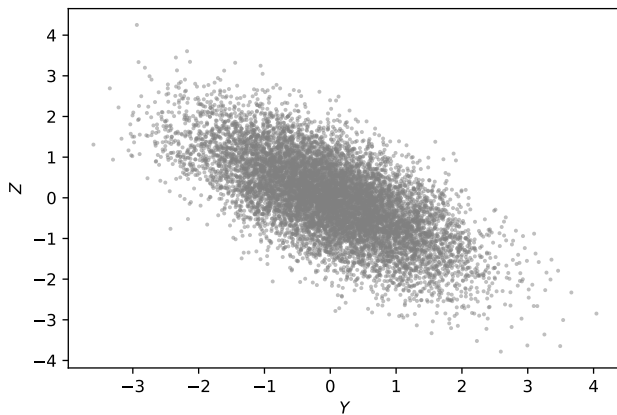
Scatterplot of (X, Y, Z)



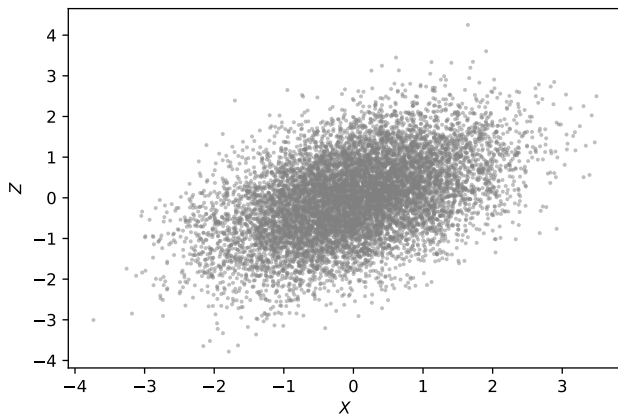
Example: Gaussian data



Example: Gaussian data

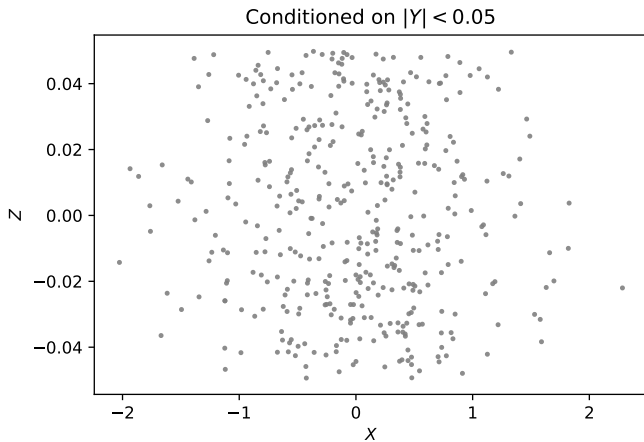


Example: Gaussian data



Example: Gaussian data

But when we condition on $Y \approx 0$:



Gaussian example

This is revealed in the “precision matrix”

$$\Omega \equiv \Sigma^{-1} = \begin{pmatrix} 1 & \frac{9}{10} & 0 \\ \frac{9}{10} & 2 & \frac{9}{10} \\ 0 & \frac{9}{10} & 1 \end{pmatrix}$$

The zeros indicate conditional independence assumptions

Undirected graphs

Simplest case:



Here $V = \{X, Y, Z\}$ and $E = \{(X, Y), (Y, Z)\}$.

This encodes the independence relation

$$X \perp\!\!\!\perp Z \mid Y$$

which means that *X and Z are independent conditioned on Y*.

Markov Property

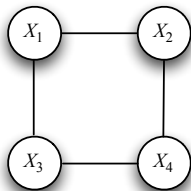
A probability distribution P satisfies the *global Markov property* with respect to a graph G if:

for any disjoint vertex subsets A , B , and C such that C separates A and B ,

$$X_A \perp\!\!\!\perp X_B \mid X_C.$$

- X_A are the random variables X_j with $j \in A$.
- C separates A and B means that there is no path from A to B that does not pass through C .

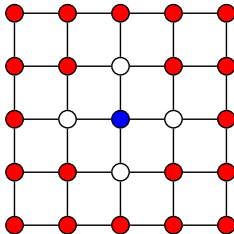
Example



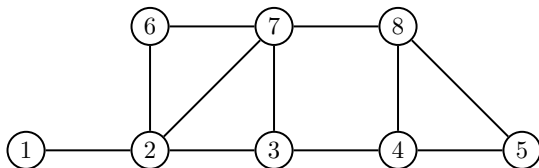
$$\begin{array}{l|l} X_1 \perp\!\!\!\perp X_4 & X_2, X_3 \\ X_2 \perp\!\!\!\perp X_3 & X_1, X_4 \end{array}$$

Example: 2-dimensional grid

The blue node is independent of the red nodes given the white nodes.



Example



$C = \{3, 7\}$ separates $A = \{1, 2\}$ and $B = \{4, 8\}$. Hence,

$$\{X_1, X_2\} \perp\!\!\!\perp \{X_4, X_8\} \quad | \quad \{X_3, X_7\}$$

Special case

If $(i, j) \notin E$ then

$$X_i \perp\!\!\!\perp X_j \mid \{X_k : k \neq i, j\}$$

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If $(i, j) \notin E$ then

$$X_i \perp\!\!\!\perp X_j \mid \{X_k : k \neq i, j\}$$

Lack of an edge from i to j implies that X_i and X_j are independent given all of the other random variables.

Graph estimation

- A graph G represents the class of distributions, $\mathcal{P}(G)$, the distributions that are Markov with respect to G
- Graph estimation: Given n samples $X_1, \dots, X_n \sim P$, estimate the graph G .

Gaussian case

Let $\Omega = \Sigma^{-1}$ be the precision matrix.

A zero in Ω indicates a *lack of the corresponding edge* in the graph

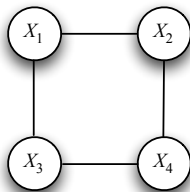
Gaussian case

$$\Omega \equiv \Sigma^{-1} = \begin{pmatrix} * & * & 0 \\ * & * & * \\ 0 & * & * \end{pmatrix}$$



Gaussian case

$$\Omega \equiv \Sigma^{-1} = \begin{pmatrix} * & * & * & 0 \\ * & * & 0 & * \\ * & 0 & * & * \\ 0 & * & * & * \end{pmatrix}$$



$$X_1 \perp\!\!\!\perp X_4 \mid X_2, X_3$$

The machine learning problem

How do we estimate the graph from a sample of data?

Gaussian case: Algorithms

Two approaches:

- parallel lasso
- graphical lasso

Parallel Lasso:

- 1 For each $j = 1, \dots, p$ (in parallel): Regress X_j on all other variables using the lasso.
- 2 Put an edge between X_i and X_j if each appears in the regression of the other.

Graphical Lasso (glasso)

- Assume a multivariate Gaussian model
- Subtract out the sample mean
- Minimize the negative log-likelihood of the data, subject to a constraint on the sum of the absolute values of the inverse covariance

Graphical Lasso (glasso)

The glasso optimizes the parameters of $\Omega = \Sigma^{-1}$ by minimizing:

$$\text{trace}(\Omega S_n) - \log |\Omega| + \lambda \sum_{j \neq k} |\Omega_{jk}|$$

where $|\Omega|$ is the determinant and S_n is the sample covariance

$$S_n = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$$

Derivation: Where does this come from?

Assume mean is zero. Then the probability density at a data point x is

$$\begin{aligned} p(x) &= \frac{1}{\sqrt{(2\pi)^p |\Sigma|}} \exp\left(-\frac{1}{2} x^T \Sigma^{-1} x\right) \\ &= \frac{1}{\sqrt{(2\pi)^p |\Sigma|}} \exp\left(-\frac{1}{2} x^T \Omega x\right) \\ &= \frac{1}{\sqrt{(2\pi)^p |\Sigma|}} \exp\left(-\frac{1}{2} \text{trace}(\Omega x x^T)\right) \end{aligned}$$

Therefore, using $\log |A| = -\log |A^{-1}|$, up to an additive constant,

$$-\log p(x) = \frac{1}{2} \log |\Sigma| + \frac{1}{2} \text{trace}(\Omega x x^T) = -\frac{1}{2} \log |\Omega| + \frac{1}{2} \text{trace}(\Omega x x^T)$$

Derivation: Where does this come from?

Summing over all the data we have

$$\begin{aligned}-\sum_{i=1}^n \log p(x_i) &= \frac{1}{2} \sum_{i=1}^n \text{trace}(\Omega x_i x_i^T) - \frac{n}{2} \log |\Omega| \\ &= \frac{n}{2} \text{trace}(\Omega S_n) - \frac{n}{2} \log |\Omega|\end{aligned}$$

Rescaling by $2/n$ and adding the ℓ_1 penalty, we get the objective function

$$\mathcal{O}(\Omega) = \text{trace}(\Omega S_n) - \log |\Omega| + \lambda \sum_{k \neq j} |\Omega_{jk}|$$

This is a convex function of Ω

Graphical Lasso (glasso)

There is a simple blockwise gradient descent algorithm for minimizing this function. It is similar to the algorithm for the lasso that we studied.

Python packages: `sklearn.covariance.GraphicalLasso` and `sklearn.covariance.GraphicalLassoCV`

Graphs on the S&P 500

- Data from Yahoo! Finance (`finance.yahoo.com`).
- Daily closing prices for 452 stocks in the S&P 500 between 2003 and 2008 (before onset of the “financial crisis”).
- Log returns $X_{tj} = \log(S_{t,j}/S_{t-1,j})$.
- Outliers capped at $\pm 6\sigma$.
- In following graphs, each node is a stock, and color indicates an industry sector

Consumer Discretionary

Energy

Health Care

Information Technology

Telecommunications Services

Consumer Staples

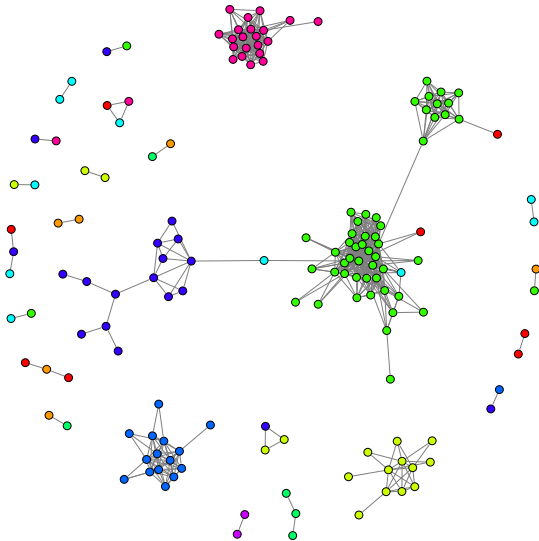
Financials

Industrials

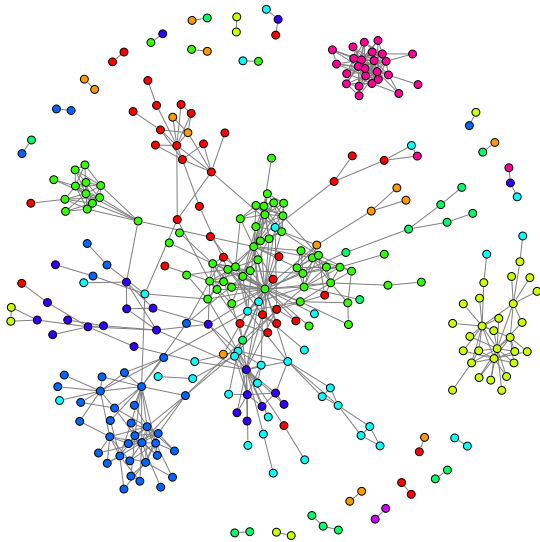
Materials

Utilities

S&P 500: Graphical Lasso



S&P 500: Parallel Lasso



Example Neighborhood

Yahoo Inc. (Information Technology):

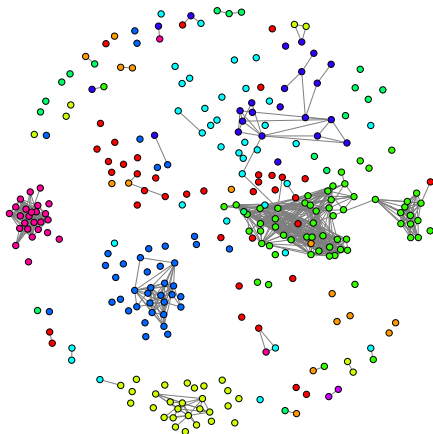
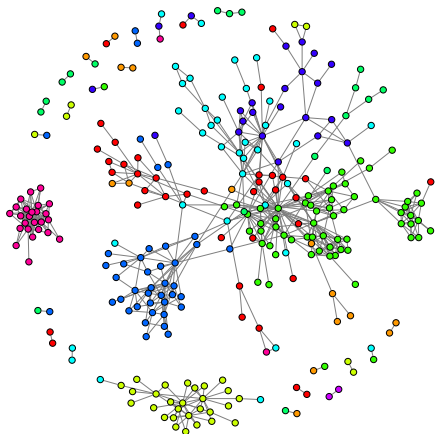
- Amazon.com Inc. (Consumer Discretionary)
- eBay Inc. (Information Technology)
- NetApp (Information Technology)

Example Neighborhood

Target Corp. (Consumer Discretionary):

- Big Lots, Inc. (Consumer Discretionary)
- Costco Co. (Consumer Staples)
- Family Dollar Stores (Consumer Discretionary)
- Kohl's Corp. (Consumer Discretionary)
- Lowe's Cos. (Consumer Discretionary)
- Macy's Inc. (Consumer Discretionary)
- Wal-Mart Stores (Consumer Staples)

Parallel vs. Graphical



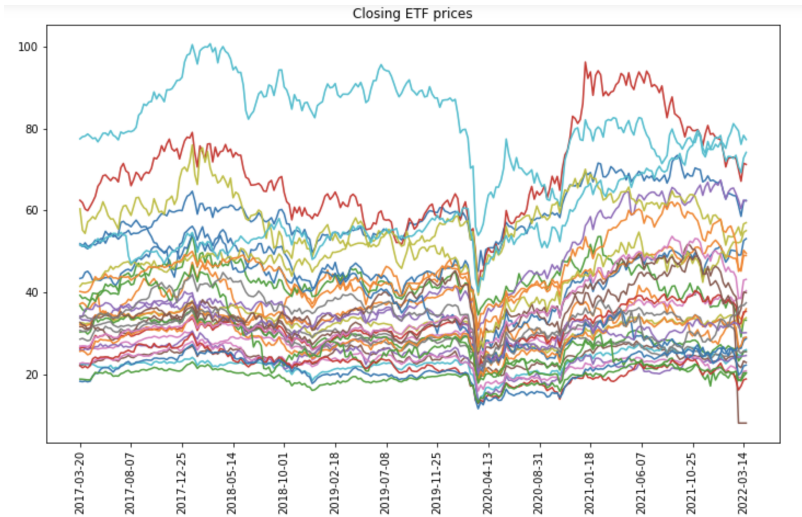
Choosing λ

Can use:

- ① Cross-validation
- ② $\text{BIC} = \text{log-likelihood} - (p/2) \log n$
- ③ $\text{AIC} = \text{log-likelihood} - p$

where p = number of parameters.

Let's go to the demo!



Summary

- Graphs encode conditional independence assumptions
- Sparse graphs represent low-dimensional structure in high dimensional data
- Gaussian case: Graph read off from precision matrix
- Graphical lasso used to estimate the graph